

Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	6aba3aa9ca93a27509644a39
Project Name	ComicCraft — AI Comic Story Creator
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

ComicCraft uses a modular FastAPI backend with Jinja2/HTML frontend integration. The presentation shows separate service modules for Gemini Flash and comic generation, with Gemini Pro and Stable Diffusion supporting narration and illustrations.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	The service code is presented in a clear, structured format.	Yes	Readable function structure is shown.
2	Proper File Structure	Services are separated into modules such as <code>gemini_flash.py</code> and <code>comic_service.py</code> .	Yes	Modular service layout is demonstrated.
3	Meaningful Variable Names	Names such as <code>prompt</code> , <code>response</code> , <code>story</code> , <code>title</code> and <code>panels</code> clearly indicate purpose.	Yes	Variables match their roles in the pipeline.
4	Function / Method Names	Functions such as <code>generate_story()</code> and <code>generate_comic()</code> are descriptive.	Yes	Function names clearly describe actions.

Code-Layout, Readability and Reusability — Continued

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
5	Code Comments	The presentation includes code structure and explanatory descriptions for the services.	Partial	Documentation is shown, but extensive inline comments are not demonstrated.
6	Modular Design	Story generation and comic generation are separated into reusable service functions.	Yes	Supports future extension and reuse.
7	No Redundant Code	The shown service flow uses focused functions for separate processing stages.	Yes	No obvious duplicate logic is shown in the presented code.
8	Error Handling	The supplied presentation does not show explicit exception-handling code.	Partial	Error-handling implementation is not demonstrated in the presentation.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	generate_story()	Python + Gemini Flash	Comic story outline generation	High
2	generate_comic()	Python	Comic generation service pipeline	High
3	improve_story()	Python + Gemini Pro	Narration and dialogue expansion	High
4	Stable Diffusion pipeline	Diffusers / Stable Diffusion	Comic illustration generation	High
5	FastAPI + Jinja2	Python / HTML / Jinja2	Frontend-to-backend application flow	High

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Modular FastAPI service structure is clearly presented.
Readability	4	Function and variable names are meaningful and the workflow is clearly documented.
Reusability	5	Separate story, comic, narration and image-generation components support reuse.
Documentation / Comments	3	The presentation documents the modules, but extensive inline comments are not shown.
Overall Score	4.25 / 5	Strong modular design with clear AI pipeline separation.

Project Details from the Presentation:

Item	ComicCraft Details
Application	Web-based AI comic story creator
Backend	FastAPI
Frontend	HTML + CSS + Jinja2
Story AI	Gemini Flash + Gemini Pro
Image AI	Stable Diffusion via Diffusers
Inputs	Story prompt, character, setting, tone, art style
Outputs	Panel-by-panel comic and PDF export
Workflow	Prompt → 5-panel outline → narration/dialogue → image prompts → illustrations → layout → PDF

Team Members:

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